Multi-hop D2D Communication

✔️ Definition

Multi-hop D2D communication is a technique where **data is transmitted from source to destination through multiple intermediate devices (relay nodes)** instead of direct communication.

- Data is transmitted through **multiple intermediate devices**
- Each device acts as a **relay node**

✔️ Basic Idea

Instead of : 📱 → 📱 (direct)

Use : 📱 → 📱 → 📱 → 📱

Instead of : A → E
Use : A → B → C → D → E

✔️ Need for Multi-hop

- Devices are far apart
- No base station available
- Signal cannot reach directly
- Direct communication not possible
- Signal strength is weak

✔️ Working Principle

1. Source device sends data to nearest device
2. That device acts as a relay
3. Data is forwarded step-by-step
4. Finally reaches destination

- Each device acts as:
    - Transmitter
    - Receiver
    - Relay

✔️ Advantages

- Improves connectivity
- Useful in disaster situations
- Better coverage
- Reduces transmission power
- Flexible communication

✔️ Disadvantages

- Delay increases (more hops)
- Battery usage increases
- Routing becomes complex

✔️ Example

- In a forest or rural area:
    - No tower
    - Devices pass messages one by one
